

Department of Information Systems and Technologies

CTIS151 – Introduction to Programming

FALL 2026 - 2027

Lab Guide #4 - Week 4 - 1

OBJECTIVES : - Solving simple arithmetic problems that require usage of input values.
- Displaying the result of a simple calculation, if and if... else statement!

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Q1. Write a C program that takes 2 double values (**a, b**) and 2 integer values (**c, d**) and performs the following mathematical equation.

$$\left| \frac{\sqrt{a} + b^2}{\frac{c^2}{\sqrt{3^d}} - \frac{a^3}{5} + 4} \right|$$

- Check your equation if all variables are set to 1. (Result = 0.4569)

Example Run #1:

Enter a and b: 4.0 1.5

Enter c and d: 2 4

Result is: 0.5086

Example Run #2:

Enter a and b: 2.5 3.0

Enter c and d: 3 2

Result is: 2.7306

Project Name: LG04_Q1

File Name: Q1.cpp

Q2. Write a C program that will read two integer numbers from the user, do the following operation, and display an appropriate message (0 means the result is false, and 1 means it is true).

$$|n1 + n2| + 5 = 25 \text{ OR } (n1/n2 \neq 3) \text{ AND } (n1 + n1 \times n2 - 2)' < 1$$

Example Run #1:

Enter the first number: 15

Enter the second number: 5

The result is 1

Example Run #2:

Enter the first number: 6

Enter the second number: 2

The result is 0

Project Name: LG04_Q2

File Name: Q2.cpp

Q3. Write a program that reads the lengths of two sides of a triangle and prints whether the triangle is isosceles or a scalene.

Project Name: LG04_Q3

File Name: Q3.cpp

Example Run #1:

Enter the first side: 7.5

Enter the second side: 7.5

It is an ISOSCELES TRIANGLE!

Example Run #2:

Enter the first side: 6.2

Enter the second side: 9.4

It is a SCALENE TRIANGLE!

Q4. An online shopping site offers free shipping for orders over 300 TL. Otherwise, a fixed shipping fee of 45 TL is added to the total.

Write a C program that reads the order amount from the user and calculates the total payment amount. If the order amount is above 300, the program applies free shipping and displays the message "You earned FREE SHIPPING :)". Otherwise, displays the message "Shipping fee is added :(". The program should also display the payment amount.

Example Run #1:

```
Enter total order amount: 450
```

```
You earned FREE SHIPPING :)
Please pay: 450.00 TL
```

Example Run #2:

```
Enter total order amount: 180
```

```
Shipping fee is added :(
Please pay: 225.00 TL
```

Project Name: LG04_Q4

File Name: Q4.cpp

Q5. In an electric guitar workshop audition, a candidate is evaluated based on four distinct performance criteria: **Rhythm, Lead Solo, Technique, and Improvisation**. The program calculates the candidate's total performance score and determines whether they are eligible for studio recording based on specific weighted threshold conditions.

Write a C program that gets the rhythm, the lead solo, the technique, and the improvisation scores as input from the user and calculates and displays the total performance score using the following percentages:

- 10% of the rhythm score,
- 20% of the lead solo score,
- 40% of the technique score,
- 30% of the improvisation score, displays a message about whether the guitarist is eligible or not for the studio recording:

If the guitarist gets at least **20** points from the Rhythm plus the Lead Solo weighted scores and **at least 45** points from the Technique plus the Improvisation weighted scores, the guitarist will be **eligible** for the Studio Recording.

Example Run1:

```
Enter rhythm score      : 80
Enter lead solo score   : 75
Enter technique score   : 85
Enter improvisation score: 70
```

```
Total performance score is 78.00
```

```
The guitarist is eligible for studio recording.
```

Example Run#2:

```
Enter rhythm score      : 60
Enter lead solo score   : 50
Enter technique score   : 65
Enter improvisation score: 40
```

```
Total performance score is 54.00
```

```
The guitarist is NOT eligible for studio recording.
```

Project Name: LG04_Q5

File Name: Q5.cpp

Q6. Write a C program that gets three values for the variables x, y, and z of a mathematical equation to compute the result of the equation given to you below, using built-in functions. Your program must also check for any occurrence of division by zero, as well as values inside the root functions, as they cannot be negative.

$$\frac{\sqrt{xy^3} - \frac{z^2}{x}}{\frac{3y}{\sqrt[3]{x+z}}} - |2x - y|$$

- Check your equation if all variables are set to 1. (Result = -2.587)

Example Run#1:

Enter the values of x, y and z respectively: 0 5 2

There is an error caused by the values.

Example Run#2:

Enter the values of x, y, and z respectively: -4 7 4

There is an error caused by the values.

Example Run#3:

Enter the values of x, y and z respectively: 7.61 0.51 2.2

The result is -14.695

Project Name: LG04_Q6
File Name: Q6.cpp

Additional Questions

AQ1.

Write a C program that calculates attendance obligation for the given lectures in a department. There are two types of attendance. Obligation, which is 75% for most courses and 50% for elective courses. Program gets;

- Number of weeks in a department,
- Number of lecture hours in a week,
- Lecture type M/m for Must, E/e for Elective,

from the user, then displays the hours of attendance obligation for the given lecture, as shown in the example run below.

Example Run#1:

```
Enter the number of weeks in a semester: 15
Enter the number of lecture hours in a week: 3
Enter the type of lecture, (M/m for Must) / (E/e for Elective): M

You must attend 34 hours
```

Example Run#2:

```
Enter the number of weeks in a semester: 14
Enter the number of lecture hours in a week: 3
Enter the type of lecture, (M/m for Must) / (E/e for Elective): E

You must attend 21 hours
```

Project Name: LG04_AQ1

File Name: AQ1.cpp

AQ2.

Write a C program that reads a five-digit binary number, converts that number to decimal, then displays the decimal number, the digits of the decimal number, and the sum of its digits as in the example run.

HINT:

$$\begin{array}{ccccccc} & 1 & & 0 & & 1 & & 0 & & 1 \\ & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ 1 * 2^4 & + & 0 * 2^3 & + & 1 * 2^2 & + & 0 * 2^1 & + & 1 * 2^0 & = & 16 + 0 + 4 + 0 + 1 = 21 \end{array}$$

Example Run#1:

```
Enter a binary number: 11011
Decimal equivalent : 27
Digits of the decimal number : 2 and 7
Sum of its digits : 9
```

Example Run#2:

```
Enter a binary number: 10111
Decimal equivalent : 23
Digits of the decimal number : 2 and 3
Sum of its digits : 5
```

Project Name: LG04_AQ2

File Name: AQ2.cpp